

Haridhanush Ravichandran

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Gender: Male **Date of birth**: 26/09/2001 **Place of birth**: Pattukkottai,Tamilnadu, India **Nationality**: Indian

ABOUT ME

Master’s student in Data Science with a strong foundation in machine learning, statistics, and data analysis. Experienced in developing end-to-end ML pipelines through academic projects using Python and PyTorch. Seeking an internship or entry-level role to apply analytical and modeling skills to real-world problems.

EDUCATION AND TRAINING

[01/10/2024 – Current]

Masters of Science (MS) in Data Science

University of Padova

City: Padova | **Country**: Italy | | **Level in EQF**: EQF level 7

[01/06/2019 – 20/05/2023]

Bachelor of Engineering in Computer Science and Engineering

Anna University

City: Tiruchirapalli | **Country**: India | | **Level in EQF**: EQF level 6

PROJECTS

[01/05/2025 – 19/06/2025]

Neuro-Symbolic Learning for MNIST Digit Relationships

Technologies: Python, PyTorch, torchvision, NumPy, Matplotlib, Machine Learning, Deep Learning, d-DNNF, Weighted Model Counting (WMC)

- Designed and trained a **Convolutional Neural Network (CNN)** for MNIST digit classification.
- Implemented **symbolic logic constraints** (e.g. $x - y \leq 1$, $(x + y) \bmod 10 = z$) using **d-DNNF representations**.
- Developed **custom PyTorch datasets** for digit pairs, triplets, and sequences with **constraint-satisfaction labels**.
- Integrated neural network predictions with **symbolic reasoning via differentiable Weighted Model Counting (WMC)**.
- Evaluated the model’s ability to **jointly perform classification and logical reasoning**.
- Visualized results including **probability distributions, WMC scores, and constraint satisfaction outcomes**.

Link: <https://haridhanush-ravichandran.github.io/projects/nesy>

[29/04/2025 – 20/06/2025]

Weather Forecasting with Transformer Models (Deep Learning Challenge)

Technologies: Python, PyTorch, NumPy, Matplotlib, Time Series Analysis, Deep Learning, Transformers

- Implemented a **data preprocessing and standardization pipeline** for **multi-station time-series weather data**.
- Designed and trained a **Transformer-based deep learning model** to **forecast future weather conditions**.
- Applied **training optimization techniques**, including **early stopping** and **learning rate scheduling**, to improve **model stability and generalization**.
- Evaluated forecasting performance using **Mean Absolute Scaled Error (MASE)** and other validation metrics.

Link: <https://haridhanush-ravichandran.github.io/projects/weather-forecast/>

Technologies: Python, PyTorch, torchvision, scikit-learn, Deep Learning, Computer Vision

- Designed and trained a **Convolutional Neural Network (CNN)** to **classify images into 20 categories** for a university deep learning challenge.
- Implemented **custom data augmentation strategies** to improve **model generalization and robustness**.
- Achieved **75.8% validation accuracy** and submitted predictions to the **challenge leaderboard**.
- Built end-to-end training and evaluation pipelines, demonstrating **proficiency in PyTorch-based deep learning workflows**.

Link: <https://github.com/Haridhanush-Ravichandran/CNN-image-classification>

TECHNICAL SKILLS

Programming

Python, R, SQL, C, C++

ML/DL Frameworks

PyTorch, TensorFlow, scikit-learn, NumPy, Pandas

Data Tools

Jupyter, Google Colab, Git, Docker, Excel, Power BI, Tableau

Data Analysis Statistics

Regression, Hypothesis Testing, Probability, Statistical Modeling, Data Cleaning, Preprocessing

CERTIFICATIONS

[Coursera, 14/07/2024]

Google Advanced Data Analytics

- Explored the roles of data professionals within an organization
- Created data visualizations and apply statistical methods to investigate data
- Built regression and machine learning models to analyze and interpret data

Mode of learning: Online

INTERNSHIPS

Accenture Virtual Internship (Data Analytics)

Analyzed datasets with SQL and Excel to identify business trends.
Developed predictive models and provided insights to support business strategy.

PwC Virtual Internship (Data Visualization)

Created business dashboards and communicated insights from data using Power BI .
Translated data into actionable recommendations for decision-making.